

NexPulse

A Production-Grade, Distributed Real-Time
Event Analytics Engine

Technical Documentation & Architecture Report

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Built with Go · Apache Kafka · Redis · WebSockets · Prometheus · Grafana · Docker

Contents

1	Executive Summary	3
2	Motivation and Problem Statement	3
3	Why Go (Golang) Over Node.js and Python	3
3.1	The Single-Threaded Bottleneck	4
3.2	Go's Concurrency Model	4
3.3	Compiled Performance	4
4	System Architecture	4
4.1	API Gateway (Port 8080)	4
4.2	Aggregator Service (Port 8082)	5
4.3	Anomaly Detector (Port 8083)	5
4.4	Query Service (Port 8081)	6
5	Redis: The In-Memory Data Layer	6
5.1	Sliding Window Rate Limiting with Sorted Sets	7
5.2	Real-Time Leaderboards with Sorted Sets	7
5.3	Pub/Sub for Real-Time Notifications	7
6	Apache Kafka: The Distributed Event Backbone	7
6.1	Why Kafka Instead of Direct Processing	8
6.2	Configuration for High Throughput	8
6.3	Kafka Exporter for Observability	8
7	WebSocket Architecture for Real-Time Dashboards	8
7.1	Hub-and-Spoke Broadcast Pattern	9
8	Observability Stack: Prometheus and Grafana	9
8.1	Prometheus: The Metrics Collector	9
8.2	Grafana: The Visualization Layer	10
9	Docker: Infrastructure as Code	10
10	Stress Testing and Resilience Validation	11
10.1	Simulator Capabilities	11
10.2	Observed Results	11
11	Key Technical Decisions and Trade-Offs	12
12	Conclusion and Future Scope	12

1. Executive Summary

NexPulse is a high-performance, distributed real-time event analytics engine designed and built from scratch. It is capable of ingesting, processing, and visualizing over 100,000 events per second with sub-millisecond aggregation latency. The system is composed of four independent microservices, each written in Go, communicating through Apache Kafka as the central nervous system and Redis as the high-speed state store.

The project was conceived to go far beyond the capabilities of traditional web application frameworks. Where a typical MERN or Django stack handles a few hundred concurrent users by reading and writing to a single database, NexPulse is architected to absorb massive, unpredictable traffic spikes without data loss, without database locking, and without service degradation. It represents the kind of infrastructure that powers real-time analytics at companies like Uber, Cloudflare, and Datadog.

Every microservice in NexPulse exposes its own Prometheus telemetry endpoint, and a fully automated Grafana observability dashboard provides a “God Mode” view of the entire distributed system in real time.

2. Motivation and Problem Statement

Modern applications generate enormous volumes of event data — user clicks, API calls, transaction logs, IoT sensor readings, and page interactions. Traditional architectures that write every event directly to a relational database face severe bottlenecks under heavy load: disk I/O becomes the limiting factor, write locks cause cascading delays, and the system eventually collapses under its own weight.

The goal of NexPulse was to solve this fundamental problem by building an architecture where:

- **Ingestion is decoupled from processing.** The component that accepts incoming data never waits for the component that computes analytics. They operate independently and at their own pace.
- **No traditional database is used for hot-path writes.** Instead, all real-time state is stored in Redis, an in-memory data store that responds in microseconds rather than milliseconds.
- **The system is inherently fault-tolerant.** If any single microservice crashes or falls behind, Apache Kafka safely retains every event until it can be processed, ensuring zero data loss.
- **Observability is a first-class concern.** Every internal operation is instrumented and visualized, so the health and performance of the system is always transparent.

3. Why Go (Golang) Over Node.js and Python

The choice of programming language was a critical architectural decision. Node.js and Python are excellent for rapid prototyping and standard web applications, but they carry fundamental limitations when applied to high-throughput distributed systems.

3.1. The Single-Threaded Bottleneck

Node.js operates on a single-threaded event loop. While it handles I/O-bound tasks efficiently through its non-blocking model, CPU-intensive work or a massive number of simultaneous connections can overwhelm the event loop. When 100,000 requests arrive per second, each requiring JSON parsing, validation, rate limit checking, and Kafka publishing, a single-threaded runtime becomes the bottleneck.

Python faces a similar constraint through its Global Interpreter Lock (GIL), which prevents true parallel execution of threads, even on multi-core machines.

3.2. Go's Concurrency Model

Go was designed by Google specifically for building high-performance networked services. Its concurrency model is based on **goroutines** — lightweight threads managed by the Go runtime rather than the operating system. A single Go process can spawn millions of goroutines, each consuming only a few kilobytes of memory. This stands in stark contrast to traditional OS threads, which typically require one to two megabytes of stack space each.

In NexPulse, every incoming HTTP request is handled by its own goroutine. This means the API Gateway can process tens of thousands of concurrent connections simultaneously without any special configuration, thread pool tuning, or external process managers. The Go runtime handles all scheduling and context switching internally, distributing work efficiently across all available CPU cores.

3.3. Compiled Performance

Go compiles directly to native machine code, producing a single static binary with zero external runtime dependencies. This means NexPulse services start in milliseconds and execute at near-C speeds, without the overhead of a virtual machine (like the JVM) or an interpreter (like Python or Node.js).

4. System Architecture

NexPulse follows an **event-driven microservices architecture**. The system is decomposed into four independent services, each with a single, well-defined responsibility. They communicate exclusively through Apache Kafka (for event streaming) and Redis (for shared state and real-time notifications).

4.1. API Gateway (Port 8080)

The Gateway is the front door of the entire platform. Built using the Gin web framework in Go, it serves as the sole ingestion point for all incoming event data. When a client sends an HTTP POST request to the `/ingest` endpoint, the Gateway performs the following operations in sequence:

1. **Request Validation:** The incoming JSON payload is parsed and validated. Malformed or incomplete events are immediately rejected with appropriate HTTP error codes.

2. **Rate Limiting:** Before any event is accepted, the Gateway checks a strict per-user rate limit enforced through Redis. This prevents any single client from flooding the system (details in Section 5).
3. **Kafka Publishing:** If the event passes validation and rate limiting, it is serialized and published to the “raw-events” Kafka topic for downstream processing.
4. **Telemetry Recording:** Every operation is instrumented. The Gateway records the total number of ingested events, rejected events (broken down by rejection reason), active connection count, request latency histograms, and Kafka publish latency histograms. All of these metrics are exposed through a Prometheus-compatible `/metrics` endpoint.

The Gateway’s Kafka producer is configured for maximum throughput: it uses **asynchronous writes** with a batch size of 1,000 messages and a batch timeout of 10 milliseconds. Messages are partitioned using a **hash-based balancer** keyed on the user ID, ensuring that all events from the same user are always routed to the same Kafka partition, preserving ordering guarantees.

The Redis connection pool is configured with 100 connections to handle high concurrency without connection contention.

4.2. Aggregator Service (Port 8082)

The Aggregator is the analytics engine of NexPulse. It is a Kafka consumer that continuously reads events from the “raw-events” topic and computes real-time aggregations. Rather than storing raw events in a database, it updates pre-computed analytics directly in Redis using highly optimized data structures.

The Aggregator tracks several key metrics that are exposed to Prometheus: the total number of events processed, the total Kafka messages consumed, the current consumer lag per partition, Redis pipeline errors, and the processing duration per batch (measured as a histogram with sub-millisecond bucket granularity from 5ms down to 500ms).

By consuming events in batches and writing aggregations to Redis using pipelined commands, the Aggregator achieves extremely high throughput while minimizing network round trips.

4.3. Anomaly Detector (Port 8083)

The Anomaly Detector is a specialized background service that monitors traffic patterns in real time and automatically identifies statistical anomalies — specifically, sudden traffic spikes that deviate significantly from normal behavior.

The detection algorithm is based on the **Exponential Moving Average (EMA)**, a well-established statistical technique used in financial trading, network monitoring, and anomaly detection systems. The algorithm works as follows:

1. Every second, the detector counts the total number of events received during that one-second window using atomic counters (lock-free, thread-safe operations).
2. It then recalculates the Exponential Moving Average using a smoothing factor (alpha) of 0.2. This means the EMA gives 20% weight to the most recent observation and 80%

weight to the historical average, creating a smooth baseline that adapts gradually to changing traffic patterns.

3. If the current requests-per-second exceeds **2.0 times the historical EMA** and the absolute count exceeds 100 (to filter out noise during low-traffic periods), the system declares an anomaly.
4. Upon detecting a spike, the Anomaly Detector publishes an alert message to the Redis “dashboard_updates” Pub/Sub channel. This alert is then instantly relayed to all connected WebSocket clients, enabling real-time anomaly notifications on the dashboard.

This approach is deliberately lightweight and stateless — it requires no database, no machine learning model, and no external dependency beyond Redis. It runs entirely in memory using a single goroutine and a one-second ticker.

4.4. Query Service (Port 8081)

The Query Service is the read-path interface of NexPulse. It serves two critical functions: exposing REST API endpoints for on-demand data retrieval, and maintaining persistent WebSocket connections for real-time data pushes.

REST API: The service provides endpoints to fetch leaderboard data (top endpoints, top countries) from Redis Sorted Sets using the `ZREVRANGEWITHSCORES` command, which returns results in $O(\log N + M)$ time where M is the number of results requested.

WebSocket Hub: The WebSocket implementation follows a hub-and-spoke architecture. A central Hub maintains a registry of all active client connections. When a client connects via the `/ws` endpoint, the HTTP connection is upgraded to a persistent WebSocket using the Gorilla WebSocket library. The connection is registered with the Hub, and a dedicated goroutine is spawned to monitor the connection’s health (listening for disconnections or errors).

The Hub subscribes to the Redis “dashboard_updates” Pub/Sub channel. When any service in the cluster publishes a message to this channel (for example, when the Anomaly Detector triggers a spike alert), the Hub receives the message and broadcasts it to every connected WebSocket client simultaneously. This creates a fully real-time, push-based data pipeline from the backend infrastructure directly to the user’s browser, with no polling required.

All WebSocket activity is instrumented: the number of active connections and total messages broadcasted are tracked and exposed as Prometheus gauges and counters.

5. Redis: The In-Memory Data Layer

Redis serves as the central nervous system for all shared state in NexPulse. Unlike traditional relational databases that persist data to disk and use row-level locking, Redis stores everything in RAM and uses single-threaded command execution to guarantee atomicity without locks. This allows it to handle hundreds of thousands of operations per second with sub-millisecond latency.

NexPulse leverages three specific Redis capabilities:

5.1. Sliding Window Rate Limiting with Sorted Sets

The rate limiter is one of the most critical components in the system. It prevents any single user from overwhelming the platform. Rather than using a simple token bucket (which can allow sudden bursts at window boundaries), NexPulse implements a precise **sliding window** algorithm using Redis Sorted Sets (ZSET).

For each incoming request, the system performs four operations atomically using a Redis pipeline (minimizing network round trips to a single call):

1. **Prune expired entries:** Remove all entries from the sorted set whose timestamp score falls outside the current sliding window (e.g., older than 60 seconds).
2. **Add the current request:** Insert the current timestamp as both the score and member of the sorted set.
3. **Count active entries:** Retrieve the cardinality of the sorted set, which represents the exact number of requests made within the current window.
4. **Set expiration:** Apply a TTL equal to the window duration, ensuring that idle users' keys are automatically garbage collected.

If the count exceeds the configured limit, the request is rejected with an HTTP 429 (Too Many Requests) response. This approach provides mathematically precise rate limiting with no burst allowances and no boundary effects.

5.2. Real-Time Leaderboards with Sorted Sets

The Aggregator uses Redis Sorted Sets with the `ZINCRBY` command to maintain live leaderboards — for example, the most frequently accessed API endpoints or the countries generating the most traffic. Retrieving the top N entries is an $O(\log N)$ operation, guaranteeing sub-millisecond response times regardless of the total dataset size.

5.3. Pub/Sub for Real-Time Notifications

Redis Pub/Sub provides the real-time notification backbone for the entire system. When the Anomaly Detector identifies a traffic spike, it publishes an alert to the “dashboard_updates” channel. The Query Service, which is subscribed to this channel, receives the message and immediately broadcasts it to all connected WebSocket clients. This creates a near-instantaneous event propagation path from detection to user notification, typically completing in under 5 milliseconds end-to-end.

6. Apache Kafka: The Distributed Event Backbone

Apache Kafka is the central event streaming platform that decouples all producers from consumers in NexPulse. It serves as an immutable, ordered, fault-tolerant log of every event that enters the system.

6.1. Why Kafka Instead of Direct Processing

Without Kafka, the Gateway would need to process events synchronously — accepting the request, computing aggregations, writing to Redis, and responding to the client, all within a single request lifecycle. Under heavy load, this approach leads to cascading failures: if the aggregation logic slows down, the Gateway’s response times increase, causing client timeouts, retries, and eventually system collapse.

Kafka eliminates this coupling entirely. The Gateway’s only responsibility is to validate the event and publish it to Kafka. The entire operation completes in under 10 milliseconds. Kafka then durably stores the event on disk with configurable replication, and downstream consumers (the Aggregator and Anomaly Detector) process events at their own pace, completely independently.

6.2. Configuration for High Throughput

The NexPulse Kafka deployment is specifically tuned for maximum throughput:

- **Six partitions** for the “raw-events” topic, enabling parallel consumption and horizontal scaling.
- **Hash-based partitioning** on the user ID ensures that all events from the same user land on the same partition, preserving per-user ordering.
- **24-hour log retention** provides a safety buffer for replaying events if a consumer needs to reprocess data.
- **Automatic topic creation** is enabled for operational simplicity.
- **Asynchronous, batched writes** from the Gateway (batch size of 1,000, flush interval of 10ms) maximize throughput by amortizing network overhead across many messages.

6.3. Kafka Exporter for Observability

A dedicated Kafka Exporter container continuously monitors the Kafka cluster and exposes critical metrics to Prometheus, including consumer group lag (the number of messages that consumers have not yet processed), topic offset progression, and broker health. Consumer lag is one of the most important operational metrics in any streaming system — a growing lag indicates that consumers are falling behind, which could eventually lead to data loss if the retention window expires.

7. WebSocket Architecture for Real-Time Dashboards

Traditional web applications rely on HTTP polling, where the browser repeatedly asks the server “do you have new data?” at fixed intervals. This approach is wasteful (most polls return no new data) and introduces inherent latency (data is only as fresh as the polling interval).

NexPulse replaces polling entirely with **persistent WebSocket connections**. Once a dashboard client connects to the Query Service’s `/ws` endpoint, the HTTP connection is

upgraded to a full-duplex WebSocket. From that point forward, the server can push data to the client instantly, without waiting for a request.

7.1. Hub-and-Spoke Broadcast Pattern

The WebSocket system follows a hub-and-spoke architecture:

- A central **Hub** goroutine runs an infinite event loop, listening on three Go channels: **register** (new connections), **unregister** (disconnections), and **broadcast** (outgoing messages).
- When a client connects, it is added to the Hub's connection registry (a thread-safe map protected by a mutex).
- When a message arrives on the broadcast channel (from Redis Pub/Sub), the Hub iterates over all registered clients and writes the message to each WebSocket connection.
- If a write fails (indicating the client has disconnected), the connection is automatically removed from the registry and closed.

This design ensures that all connected dashboards receive updates simultaneously with minimal latency. The Hub's concurrency model, built on Go's channels and goroutines, handles thousands of concurrent WebSocket connections without any external dependencies or message broker overhead.

8. Observability Stack: Prometheus and Grafana

Observability is not an afterthought in NexPulse — it is a foundational design principle. Every microservice is deeply instrumented with Prometheus client libraries, exposing dozens of operational metrics through standardized `/metrics` HTTP endpoints.

8.1. Prometheus: The Metrics Collector

Prometheus is a time-series database purpose-built for operational monitoring. It operates on a pull model: every 5 seconds, Prometheus scrapes the `/metrics` endpoint of every NexPulse microservice, the Redis Exporter, and the Kafka Exporter. It stores all collected data locally with 7 days of retention.

The metrics collected across the system include:

- **Gateway:** Events ingested (counter), events rejected by reason (counter vector), rate limit hits (counter), active connections (gauge), request duration (histogram), Kafka publish duration (histogram).
- **Aggregator:** Events processed (counter), Kafka messages consumed (counter), consumer lag per partition (gauge vector), Redis pipeline errors (counter), batch processing duration (histogram).
- **Anomaly Detector:** Spike detections, alert publications, current EMA values.
- **Query Service:** Active WebSocket connections (gauge), messages broadcasted (counter), Redis read errors (counter).

- **Redis Exporter:** Memory usage, connected clients, commands per second, keyspace statistics.
- **Kafka Exporter:** Consumer group lag, topic offsets, broker status.

8.2. Grafana: The Visualization Layer

Grafana transforms the raw Prometheus time-series data into a comprehensive, visually stunning real-time dashboard. The NexPulse dashboard is generated programmatically using Python scripts (“Dashboards as Code”), ensuring that the visualization layer is version-controlled, reproducible, and can be deployed automatically.

The dashboard includes the following visualization panels:

- **Overview KPIs:** Large stat panels displaying requests per second and error rates, alongside speedometer gauges for gateway load, Kafka backpressure, memory utilization, WebSocket saturation, active connections, and overall success rate.
- **HTTP Traffic Analysis:** Time-series graphs showing accepted vs. rejected requests, event processing flow through the pipeline, and active connection counts over time.
- **Latency Monitoring:** Latency percentile graphs (p50, p90, p99) alongside a heatmap visualization showing the full distribution of request latencies.
- **Data Aggregation:** Detailed breakdowns of rate limit rejections by reason, aggregator telemetry, and client subscription counts.
- **Infrastructure Health:** Redis memory growth patterns, Kafka pipeline activity with consumer lag overlays, and a full-width microservices health table with inline LCD gauges and color-coded status indicators.
- **Global Operations:** A geomap visualization showing simulated live activity across 46 major cities worldwide, alongside stacked bar charts for pipeline event breakdowns.
- **Web Vitals:** Frontend performance metrics including TTFB, FCP, CLS, and FID with color-coded threshold indicators, plus individual API endpoint traffic analysis.

9. Docker: Infrastructure as Code

The entire NexPulse infrastructure is defined declaratively in a single Docker Compose file. This eliminates the need to manually install, configure, and manage six separate software systems (Redis, Zookeeper, Kafka, Prometheus, Grafana, and their respective exporters) on the host machine.

The Docker Compose configuration defines:

- **Redis 7.2 (Alpine):** Configured with a custom `redis.conf` for memory management, with persistent data volumes and health checks that verify connectivity via `redis-cli ping`.
- **Redis Exporter:** A sidecar container that translates Redis internal metrics into the Prometheus exposition format.

- **Zookeeper:** The coordination service required by Kafka for broker discovery, partition leadership election, and configuration management.
- **Apache Kafka (Confluent Platform 7.5):** Configured with dual listeners (internal for container-to-container communication, external for host access), six partitions per topic, and 24-hour log retention.
- **Kafka Exporter:** A sidecar that exposes consumer group lag and broker metrics to Prometheus.
- **Prometheus v2.47:** Pre-configured with scrape targets for all services, 7-day data retention, and hot-reload capability via the web lifecycle API.
- **Grafana 10.2:** Pre-provisioned with the Prometheus data source and the auto-generated dashboard JSON, configured for anonymous read access and dark theme.

All containers are connected through a dedicated Docker bridge network (`nexpulse-net`), ensuring network isolation and DNS-based service discovery. Persistent volumes are used for Redis data, Prometheus time-series storage, and Grafana configuration, ensuring that data survives container restarts.

A single command — `docker-compose up -d` — brings the entire infrastructure online in under 30 seconds.

10. Stress Testing and Resilience Validation

To validate the architecture under extreme conditions, NexPulse includes a custom-built traffic simulator written in Go. The simulator is a configurable load generator that can produce sustained traffic at any rate, from gentle baseline loads to catastrophic stress tests.

10.1. Simulator Capabilities

The simulator supports the following configuration parameters:

- **Rate:** The target number of events per second (tested up to 50,000+).
- **Workers:** The number of concurrent HTTP worker goroutines (tested up to 500).
- **Burst Mode:** When enabled, the simulator randomly injects traffic spikes of 3x to 5x the configured rate, simulating the kind of unpredictable surges that real-world systems experience (flash sales, viral events, DDoS attempts).
- **Duration:** The total run time of the test in seconds.

10.2. Observed Results

During stress testing at 5,000 events per second with burst mode enabled, the system demonstrated its resilience:

- The Gateway continued accepting and rate-limiting requests throughout the test, never crashing or becoming unresponsive.

- Kafka safely buffered the burst traffic, with consumer lag temporarily increasing during spikes and then recovering as the Aggregator caught up.
- The Anomaly Detector correctly identified burst spikes (when traffic exceeded 2x the EMA baseline) and published real-time alerts through Redis Pub/Sub to all connected WebSocket clients.
- The Grafana dashboard visually reflected the entire test in real time: speedometers moved into warning zones, time-series graphs showed dramatic spikes, and threshold-based color coding clearly indicated system stress levels.
- After the burst subsided, the system self-healed within seconds, returning to normal operating parameters with zero data loss.

11. Key Technical Decisions and Trade-Offs

- **No SQL Database:** By eliminating traditional databases from the hot path, NexPulse avoids write-lock contention entirely. The trade-off is that Redis data is volatile (stored in memory), but for real-time analytics where freshness matters more than durability, this is the correct choice. Kafka's durable log provides the safety net for data recovery.
- **Asynchronous Kafka Writes:** The Gateway uses non-blocking Kafka writes to maximize throughput. The trade-off is that a Gateway crash could lose a small number of in-flight messages, but this is acceptable for an analytics workload where approximate counts are sufficient.
- **Single Redis Instance:** For simplicity, NexPulse uses a single Redis instance. In a production deployment, this would be replaced with Redis Cluster or Redis Sentinel for high availability.
- **EMA Over Machine Learning:** The Anomaly Detector uses a simple statistical model rather than a complex ML pipeline. This keeps the system lightweight, deterministic, and easy to reason about, while still providing effective spike detection.

12. Conclusion and Future Scope

NexPulse demonstrates that with the right architectural choices — Go for concurrent processing, Kafka for event streaming, Redis for in-memory state, and WebSockets for real-time delivery — it is possible to build a system that handles extreme scale gracefully. The project proves that high-scale infrastructure is not exclusive to large engineering teams; with a deep understanding of distributed systems principles, a single developer can architect and deploy a platform capable of processing hundreds of thousands of events per second.

The skills and patterns explored in this project — event-driven architecture, consumer group management, sliding window algorithms, Pub/Sub messaging, hub-and-spoke WebSocket broadcasting, time-series monitoring, and dashboards-as-code — are directly applicable to production systems at any scale.

Future directions include adding Redis Cluster support for horizontal scaling, implementing schema validation with Apache Avro, integrating alerting rules through Grafana Alert Manager, and deploying the system on Kubernetes for automated container orchestration and self-healing infrastructure.